# Dialogue Four: Function

**Reader:** Functions are widely used in elementary mathematics.

**Author:** Yes, of course. You are familiar with numerical functions. Moreover, you have worked already with different numerical functions. Nevertheless, it will be worthwhile to dwell on the concept of the function. To begin with, what is your idea of a function?

**Reader:** As I understand it, a function is a certain correspondence between two variables, for example, between $x$ and $y$. Or rather, it is a dependence of a variable $y$ on a variable $x$.

**Author:** What do you mean by a "variable"?

**Reader:** It is a quantity which may assume different values.

**Author:** Can you explain what your understanding of the expression "a quantity assumes a value" is? What does it mean? And what are the reasons, in particular, that make a quantity to assume this or that value? Don't you feel that the very concept of a variable quantity (if you are going to use this concept) needs a definition?

**Reader:** O.K, what if I say: a function $y=f(x)$ symbolizes a dependence of $y$ on $x$, where $x$ and $y$ are numbers.